Ahan Bhatt

bhattahan@gmail.com | **linkedin.com/in/ahan-bhatt** | **github.com/AhanBhatt**

Summary

Software and AI engineer with an M.S. in Computer Science and Engineering and 4 years of professional experience developing enterprise AI systems, machine learning pipelines and backend applications. Experienced in production RAG, agentic workflows, GPU-accelerated transformer inference and AI evaluation. Skilled in Python, C++, PyTorch, CUDA, FastAPI and Kubernetes, with published research in machine learning.

Professional Experience

Artificial Intelligence Engineer August 2024 - August 2025
Growexx *Ahmedabad, Gujarat, India*

- Built a FHIR-integrated prior-authorization agent using hybrid RAG, LangGraph, MCP, deterministic evidence validation, and clinician approval workflows.
- Recorded 92.5% first-pass acceptance without missing-information follow-up, 84% automated evidence-criteria coverage, and 6.5 minutes of active staff time per authorization.
- Developed durable multi-agent commerce workflows using Temporal, integrating CRM, orders, inventory, and payments through policy-checked tools and human escalation.

Research Trainee (Data Science) January 2024 - May 2024
Space Applications Centre, Indian Space Research Organization *Ahmedabad, Gujarat, India*

- Developed an LSTM satellite clock-bias forecasting pipeline using first-order differencing, 10-minute resampling, and 7 to 31-day data windows; achieved 1.04 x 10^{-11} RMSE on the 14-day window.
- Deployed Kafka streaming, automated preprocessing, anomaly filtering, and a Next.js/React dashboard for live forecasting and model output monitoring.

Software Engineer (AI) February 2022 - January 2024
Intellytics *Ahmedabad, Gujarat, India*

- Engineered permission-aware enterprise RAG for a consulting firm with 1,000+ employees, connecting SharePoint, Confluence, Snowflake, and internal APIs with authorization-filtered retrieval and source citations.
- Developed a Neo4j-based fraud investigation platform using Kafka, Snowflake, FastAPI, bounded graph queries, and traceable transaction evidence.
- Built insurance document intelligence pipelines combining OCR, schema-validated extraction, policy retrieval, source provenance, and human review.

Research Experience

Graduate Researcher September 2025 - June 2026
AIEA Lab @ UCSC *Santa Cruz, California*

- Developed ReliaGuard Studio for human-AI reliance risk evaluation using calibrated ML models, symbolic rules, uncertainty estimates, and conformal risk thresholds.
- Integrated 43,263 human-AI interactions from 2,229 participants across five public datasets using FastAPI, PostgreSQL, Next.js, and audit workflows.

AI Researcher (Part-time) June 2024 - May 2025
Artificial Intelligence Institute, University of South Carolina *Columbia, South Carolina (Remote)*

- Evaluated TimeGPT, FPT, Time-MoE, Moirai, and Chronos against statistical and deep-learning baselines across five datasets for anomaly detection, prediction, and computational efficiency.
- Researched multimodal toxicity mitigation and preference optimization for text-to-image diffusion models, using PostgreSQL-backed experiment tracking for datasets, model outputs, and evaluation results.

Relevant Projects

CacheIR [**github.com/AhanBhatt/CacheIR**]

- Built a decoder-only transformer compiler/runtime with custom IR, transformer-specific fusion passes, CUDA/Triton kernels, and KV-cache-aware prefill/decode execution.
- Achieved 1.09x to 3.24x speedups in median end-to-end latency versus vLLM across 18 batch-1 BF16 Qwen3/DeepSeek workloads, with a 1.53x geometric-mean speedup and exact greedy-token parity.

BreakPoint [**github.com/AhanBhatt/BreakPoint**]

- Built a Python SDK/CLI that converts LLM failure traces into reproducible adversarial regression suites using Pydantic/YAML specifications, mutation operators, deterministic validation, and live judge adapters.
- Compiled 12 sourced failure traces into 12 base cases and 36 adversarial variants, producing 48 evaluation cases with JSONL/DuckDB exports and CI-ready regression packs.

Education

University of California, Santa Cruz [GPA: 3.82] Santa Cruz, California
Master of Science in Computer Science and Engineering *August 2025 - June 2026*

Indus Institute of Technology and Engineering [CGPA: 9.13/10] Ahmedabad, India
Bachelor of Technology in Computer Engineering *August 2020 - July 2024*

Publications

Google Scholar

- **Bhatt, A.**, Mehta, I., & Patidar, P. "*Leveraging LSTM for Predictive Modeling of Satellite Clock Bias.*" IEEE CDMA 2025.
- Shyalika, C., Bagga, H. K., **Bhatt, A.**, et al. "*Time Series Foundational Models: Their Role in Anomaly Detection and Prediction.*" AAAI 2025 Workshop.
- **Bhatt, A.**, Vaghela, N., & Dudhia, K. "*Generating Knowledge Graphs from Large Language Models.*" arXiv preprint.

Technical Skills

Languages:	Python, C++, SQL, JavaScript/TypeScript
Machine Learning & LLMs:	PyTorch, Transformers, scikit-learn, LangChain, LangGraph, RAG, MCP, LLM Evaluation
ML Systems & Compilers:	CUDA, Triton, Custom IRs, Operator Fusion, KV Caching, Quantization, pybind11, CMake
Backend & Data:	FastAPI, REST APIs, PostgreSQL, Snowflake, Kafka, Docker, Kubernetes, Linux, Git, CI/CD
Cloud:	AWS (EC2, S3, IAM, Lambda, SageMaker, CloudWatch)